performance computing in the field of science and engineering. This system bagged 96th position in the Top500.org list of top 500 supercomputers in the world.

Specs:
SahasraT (CRAY XC40) is the product of Cray Inc. an American supercomputer manufacturer. Cray XC40 is a product from OEM Cray. This Supercomputer consist of Intel Haswell Xeon E5-2680v3 processors, NVIDIA K40 GPU accelerators and Intel Xeon Phi 5120D coprocessors and whopping storage of 2.1 PB (10^{15} bytes). There are around 1500 processors and coprocessors and 44 GPUs to handle complex tasks in the system. The System implements Cray Aries interconnect routing using DragonFly topology. Cray's Linux environment is used as the OS for the system. SahasraT has been rated to 901.54 TFLOPS, which the highest rating amongst all supercomputers in India.

SaharasT provides service to our nation in the fields of aerospace engineering, meteorology predictions and astrological simulations. Also SaharasT is used for molecular and material research

India's fastest supercomputer

and mapping entire climate condition of the particular region via simulation. Overlapping of supernovae was simulated by the SahasraT system.

Aaditya (IBM/Lenovo System)
Indian Institute of Tropical Meteorology Pune, is the India's finest meteorological department which uses IBM X system supercomputer for research and development. System is called as Aaditya which is manufactured by IBM. This system is ranked at 116th position in Top500.org list.

Specs:
This system has Intel Xeon Haswell E5-2670 2.6 GHZ processors and total RAM storage of 15TB. There are 2384 compute nodes in the system. The System


The Aaditya supercomputer in Pune

is rated with a speed of 719.2 TFLOPS. Red Hat Enterprise Linux is used as the operating system environment. Nodes are connected through infiniband interconnect technology.

This supercomputer helps the institute to operate and provide accurate data regarding the Nation's weather conditions, simulating weather models of the country, predicting rainfall cycles for the monsoon, and air quality forecasting. Most of our country's weather forecasting is done by this system. Farmers rely on the information provided by this system on rainfall predictions and climate changes.

TIFR ColourBoson
This supercomputer is located at Tata Institute of Fundamental Research facility. This machine is also a Cray product, model name is Cray XC-30. This supercomputer is deployed at Hyderabad and the system is ranked at 145th position.

Specs:
ColourBoson consists of 4760 nodes of Intel Xeon E5-2680 processors and NVIDIA Tesla K20x GPU. Total storage of the system is 1.1 PB. Nodes are interconnected by Cray's Aries interconnect routing technology with DragonFly topology. Linux environment is used as

The TIFR ColourBoson

the OS for the system. The System can achieve a speed of 558.7 TFLOPS.

This system is used under Indian Lattice Gauge Theory Initiative program by the scientists of Tata Institute of Fundamental Research. Research and development on theoretical physics and

HOMEMADE SUPERCOMPUTER

Though many supercomputing systems in the nation have been bought from abroad, from manufacturers such as Cray Inc. from America, the Param series of supercomputers are designed and assembled in India by Center of Development for Advance Computing, Pune. When the US denied the import of supercomputers to the country, CDAC was set up in 1988. C-DAC was able to build India's first indigenously built supercomputer Param 8000 in 1991. The Param Yuva 2, CDAC's latest machine, was made in under 3 months, at the cost of a mere 16 crore rupees. To connect all Param series of supercomputers across the country, PARAMnet is used, which offers 2.5 GB/s bandwidth in full-duplex mode.

PRESENCE OF MIND

IBM's Watson supercomputer was built to develop natural responses to answering questions based on the quiz from the show Jeopardy. To tackle the questions Watson had all of the text in encyclopedia websites such as Wikipedia stored into the memory, keywords from question were joint and correlated with the content stored on the computer. Apart from winning popular TV quiz shows, Watson's cognitive capabilities proved beneficial in many fields.

Super Chef Watson
Watson analysed thousands of recipes and created a statistical map of various combination of ingredients, chemical compositions of the ingredients and figuring out new recipes from this data. Watson created 65 new recipes of its own which are available in the book called Cognitive Cooking with Chef Watson.